Capstone Project 3

Real-Time Streaming & Analytics Platform (Kinesis + MSK + ECS + S3 Data Lake)

Industry Scenario

A **Global Media Streaming Company** analyzing live user behavior.

Requirements:

- Real-time clickstream processing
- Kafka-based ingestion
- Long-term storage in S3
- Real-time dashboard microservices

Problem Statement

Build streaming pipeline:

- Producers send events to MSK (Kafka)
- Kinesis processes data
- Lambda transforms data
- Store in S3 Data Lake
- ECS microservice consumes analytics
- Route53 public dashboard

Use Cases

1. Real-time fraud detection
2. Clickstream analytics
3. Personalized recommendations

Architecture Components

- MSK
- Kinesis Data Streams
- Lambda
- S3

- ECS
- ECR
- Route53
- ElastiCache

Terraform Code

1 $3 Data Lake

```
resource "aws_s3_bucket" "data_lake" {
  bucket = "media-data-lake-prod"

  server_side_encryption_configuration {
    rule {
      apply_server_side_encryption_by_default {
        kms_master_key_id = aws_kms_key.platform_key.arn
        sse_algorithm     = "aws:kms"
      }
    }
  }
}
```

---

2 MSK Cluster

```
resource "aws_msk_cluster" "kafka_cluster" {
  cluster_name           = "media-kafka"
  kafka_version          = "3.5.1"
  number_of_broker_nodes = 2

  broker_node_group_info {
    instance_type = "kafka.t3.small"
    client_subnets = var.private_subnets
  }
}
```

---

3 Kinesis Stream

```
resource "aws_kinesis_stream" "analytics_stream" {
  name             = "analytics-stream"
  shard_count      = 1
}
```

---

4 Lambda Transformer

```
resource "aws_lambda_function" "transformer" {
  function_name = "stream-transformer"
  runtime       = "python3.9"
  handler       = "index.handler"
  role          = aws_iam_role.lambda_role.arn
  filename      = "transformer.zip"
}
```